

Zhalok Rahman

Software Engineer 2 at Datatree Technology

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LinkedIn: <https://www.linkedin.com/in/zhalok-rahman>

Github: <https://github.com/zhalok>

Video Introduction: <https://www.loom.com/share/c5ab46a4737244edafd9f052555767a1>

Skills

Golang, Python (FastAPI, Asyncio, Pytorch, Pandas), Node JS (Nest JS, Express JS, Koa), JavaScript/TypeScript (React JS, Next JS, PostgreSQL, MongoDB, Redis (Cache, Pub/Sub, Streams), RabbitMQ, WebSocket, SSE, Docker, AWS (ECS, CloudWatch, SQS, Kinesis, Lambda, OpenSearch, DynamoDB), Grafana, Prometheus, CI/CD (Github Actions, CodePipeline)

Education

Shahjalal University of Science and Technology

Bachelors of Science in Computer Science

2019-01-01 – 2024-04-01

Certifications

AWS Certified Cloud Practitioner

AWS

<https://www.credly.com/badges/201260d3-12e0-4d22-a49f-4e64d68b348e/>

Covered AWS core services and cloud concepts including EC2, S3, VPC, IAM, billing/pricing models, the shared responsibility model, and cloud architecture best practices

Fundamentals of Database Engineering

Udemy

<https://www.udemy.com/certificate/UC-afe8cbda-1420-4cca-a833-aed013e7a6e2/>

Learned about ACID, Data organization, Indexing, Partitioning, Sharding, Concurrency control, Replication, Sharding, Partitioning etc

MongoDB Query Optimization Techniques

MongoDB

<https://www.credly.com/badges/68a4e8c2-4325-41fe-9e5d-1363ddde2921/>

This digital credential validates your knowledge in optimizing query performance using indexing, query restructuring, and execution plans. It recognizes your understanding of identifying and resolving slow queries, as well as enhancing read and write operations.

Optimizing MongoDB Performance with Tuning Tools

MongoDB

<https://www.credly.com/badges/e861b7b4-028a-47d4-aed6-7e586d4d95da>

This digital credential validates your knowledge in applying performance tuning techniques in MongoDB through profiling, slow query analysis, and system monitoring. It recognizes your understanding of analyzing query performance, optimizing indexes to improve query efficiency and reduce latency, and configuring cache utilization to enhance read efficiency and reduce database load, ensuring improved database performance at scale.

Experience

Kinetik Healthcare

Software Engineer 2

2025-04-01 – Present

- Identified slow query shapes from MongoDB profiler
- Optimized MongoDB query performance by **41 percent** using covered atlas search index scan
- Reduced network content download time by **66 percent** with default query projection
- Enhanced read performance by keeping data denormalized and used event driven approach to update denormalized fields

PropertyFinder

Software Engineer

2024-10-01 – 2025-10-31

- Managed DynamoDB and Opensearch services
- Mitigated and fixed root causes of b2b and b2c database inconsistencies
- Fixed legacy data inconsistencies in production
- Resolved data migration issues
- Fixed legacy data wipe out

Personal Projects

MongoDB vs PostgreSQL benchmarking for storing documents

<https://github.com/zhalok/mongodb-bson-vs-postgres-jsonb-benchmarking>

- Collected metrics for query execution time and storage footprints of MongoDB and PostgreSQL
- Benchmarked between MongoDB and PostgreSQL at scale via load testing

Tech: Golang, JavaScript

postgrespector

<https://github.com/zhalok/postgrespector>

- Extracting PostgreSQL metrics and container metrics via Prometheus exporter
- Observed metrics under stress in Grafana

Tech: PostgreSQL, Prometheus, Docker, Grafana

Obstainer

<https://github.com/zhalok/obstainer>

- Built prometheus scraper for extracting container metrics from cgroup
- Visualized container metrics in grafana dashboard

Tech: Golang

Data Stores Python

<https://github.com/zhalok/data-stores-python>

- Compared between Single Threaded, Multi-threaded and Single-Threaded-Event-Driven architecture for in-memory data store
- Demo: <https://bit.ly/3UFJfPx>

Tech: Python, Socket, System Calls

Go Http From Scratch System Calls

<https://github.com/zhalok/go-http-from-scratch-system-calls>

- Built http protocol from sockets and systemcalls
- Handled graceful connection closing by handling OS signals
- Built HTTP request body parser using basic string libraries

Tech: Golang, Syscalls, OSSignals, Sockets